
Statistics & Probability

Week 3 Internship Report

Descriptive Statistics | Probability Distributions
Hypothesis Testing | Correlation vs Causation

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Datasets: Titanic (891 passengers) | Housing Prices (545 properties)

1. Introduction

This report presents the statistical analysis conducted on two datasets as part of Week 3 of the AnalystLab Africa Data Science Internship. The analysis covers descriptive statistics, probability distributions, hypothesis testing, and the critical distinction between correlation and causation.

Statistical thinking is foundational to data science. It enables us to move beyond simple observation and make data-driven inferences with measurable confidence.

2. Descriptive Statistics

2.1 Titanic Dataset

The Titanic dataset contains 891 passenger records with 11 features after cleaning. Key numerical variables analyzed: Age and Fare.

Measure	Age	Fare	Measure	Age	Fare
Mean	29.36	32.20	Skewness	0.51	4.79
Median	28.00	14.45	Kurtosis	0.99	33.40
Std Dev	13.02	49.69	Range	79.58	512.33
Min	0.42	0.00	IQR	13.00	23.09

Interpretation: Age is near-normally distributed (skew=0.51, mean~median). Fare is heavily right-skewed (skew=4.79) with extreme outliers. Mean Fare (32.2) is more than double the median (14.5).

2.2 Housing Dataset

The Housing dataset contains 545 property records with 13 features.

Measure	Price	Area	Measure	Price	Area
Mean	4,766,729	5,151	Skewness	1.21	1.32
Median	4,340,000	4,600	Kurtosis	1.96	2.75
Std Dev	1,870,440	2,170	Min	1,750,000	1,650
Max	13,300,000	16,200	IQR	2,310,000	2,760

Interpretation: Both Price (skew=1.21) and Area (skew=1.32) are moderately right-skewed. Mean price exceeds median by ~430,000.

3. Hypothesis Testing

Four statistical tests conducted at Alpha = 0.05. P-value < 0.05 means the result is statistically significant - not due to random chance.

Test 1: Age - Survivors vs Non-Survivors

Mean (Survived): 28.3 | Mean (Died): 30.0
T = -1.9395 | p = 0.0528

Result: NOT Significant

Test 2: Fare - 1st Class vs 3rd Class

Mean (1st): 84.2 | Mean (3rd): 13.7
T = 19.4487 | p < 0.001

Result: HIGHLY Significant

Test 3: Survival vs Gender

Female Survival: 74.2% | Male Survival: 18.9%
Chi2 = 260.72 | p < 0.001

Result: HIGHLY Significant

Test 4: Price - AC vs No AC

Mean (AC): 6,013,221 | Mean (No AC): 4,191,940
T = 11.8390 | p < 0.001

Result: HIGHLY Significant

Summary Table

#	Test Description	p-value	Result
1	Age: Survivors vs Died	0.0528	NOT Sig.
2	Fare: 1st Class vs 3rd Class	< 0.001	SIGNIFICANT
3	Survival vs Gender	< 0.001	SIGNIFICANT
4	Price: AC vs No AC	< 0.001	SIGNIFICANT

4. Correlation vs Causation

A critical principle in statistics: correlation does NOT imply causation.

Fare and Survival ($r = 0.26$)

Correlation: Higher fare correlates with higher survival.

Fallacy: Paying more increases survival chances.

Truth: Fare is a proxy for CLASS. First-class passengers were closer to lifeboats.

Bedrooms and Price ($r = 0.37$)

Correlation: More bedrooms correlates with higher prices.

Fallacy: Adding bedrooms increases house value.

Truth: Area ($r = 0.54$) is the stronger predictor. Many small bedrooms do not add value.

Age and Survival ($r = -0.06$, $p = 0.0528$)

Correlation: Very weak negative correlation with survival.

Fallacy: Age affects survival probability.

Truth: Not statistically significant. Effect is captured by gender and class.

Key Takeaway:

Always investigate confounding variables and domain context before concluding causation.

5. Conclusion

Week 3 established the statistical foundation for the machine learning work ahead:

1. Age is near-normally distributed; Fare and Price are right-skewed - this informs Week 6 feature transformations.
2. Three of four hypothesis tests were highly significant - gender, class, and AC are meaningful predictors.
3. Age-Survival was NOT significant ($p=0.0528$) - not every variable is useful for prediction.
4. Correlation $\neq$ Causation - domain knowledge is essential for correct interpretation.

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